Adjacency-Free Patterns on a *Physical* $3 \times 3 \times 3$ Cube: Reachability, Algorithms, and a Corrected Count

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Abstract

Our first two notes treated the cube as a free stickering: six colors placed on the faces however one likes. That is the right model for a pure combinatorics question, but it is the *wrong* model for a real twisty puzzle, where stickers are glued to *cubies*: the six centers are mutually fixed, each edge carries a fixed unordered pair of colors, and each corner a fixed triple. A pattern "exists" on a physical cube only if some legal turn sequence reaches it from the solved state. We re-examine the adjacency-free question (no two orthogonally adjacent same-face stickers) under this *reachability* model, with every claim verified by an exhaustive or exact computer search against a from-scratch cube simulator. Two findings stand out. **(1)** In the half-turn subgroup $\langle U^2, D^2, F^2, B^2, L^2, R^2 \rangle$ (order 663 552, fully enumerated) there is *exactly one* adjacency-free state: the checkerboard *Pons Asinorum*, reached by $U^2D^2F^2B^2L^2R^2$. So among half-turn patterns the configuration is *unique* — vindicating the "only one way" intuition in that regime. **(2)** In the full group the pattern is *not* unique: the *superflip* (reachable in 20 moves) is a second, genuinely different adjacency-free state, and a random walk exhibits dozens more. We also issue an **erratum** to the companion counting note: its "$\approx 1.2 \times 10^9$" figure counts *free* stickerings, the vast majority of which are *not* reachable on a real cube; in particular its "cyclic" C_6 checkerboard is physically impossible because centers are fixed.

1 Why the model must change

A free stickering ignores the mechanics of the puzzle. On a real $3 \times 3 \times 3$:

- the six **center** facelets never move relative to one another, so the color of each face is pinned (White opposite Yellow, Green opposite Blue, Red opposite Orange in the Western scheme);
- each of the 12 **edge** cubies carries a fixed *pair* of colors, and each of the 8 **corner** cubies a fixed *triple*; a legal move permutes and reorients whole cubies, it cannot repaint a sticker.

Consequently the reachable states form the Rubik group of order

$$|G_{\text{Rubik}}| = \frac{8! \cdot 3^8 \cdot 12! \cdot 2^{12}}{12} = 43\,252\,003\,274\,489\,856\,000 \approx 4.3 \times 10^{19} \quad [1, 2].$$

A pattern is *realizable* iff it is one of these states. This is the model a cuber means by "a pattern you can make," and the one the user asked for: every illustration below is annotated with an explicit algorithm that *provably* produces it (checked on the simulator described in §2).

Definition 1 (Adjacency-free / the property)**.** A cube state is *adjacency-free* if, on every face, no two orthogonally adjacent facelets (left/right/up/down within that face) share a color. The "no two same-colored pieces next to each other" property of the earlier notes.

2 Method and verification

We wrote an independent facelet-permutation simulator (54 stickers; faces ordered U, R, F, D, L, B) implementing the six quarter-turns as permutations and composing all 18 face moves from them. It passed standard identities ($X^4 = \mathrm{id}$ for each face X; the "sexy move" $(RUR'U')$ has order 6; the published superflip algorithm flips exactly the edges). Using it we (i) *exhaustively* enumerated the half-turn subgroup by breadth-first search over its 663 552 states, recording a shortest half-turn word for each, and (ii) explored the full group by a $4{\times}10^6$-step random walk. All numerical claims below are outputs of that program.

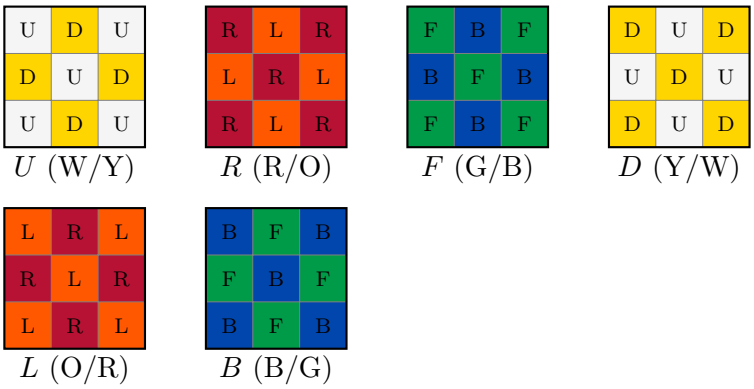
3 The half-turn subgroup: the pattern is unique

Theorem 1 (Uniqueness among half turns, verified exhaustively)**.** *Let $H = \langle U^2, D^2, F^2, B^2, L^2, R^2\rangle$, the half-turn subgroup, of order 663 552. Exactly one state of H is adjacency-free, namely the checkerboard* ***Pons Asinorum***

$$\boxed{U^2\,D^2\,F^2\,B^2\,L^2\,R^2} \qquad (\textit{equivalently } M^2E^2S^2),$$

in which every face shows its own color interleaved with its opposite *color. In particular, within the half-turn regime there is literally "only one way."*

Proof (exhaustive computation). Breadth-first search generates all 663 552 elements of H (matching the known order $|H| = 663\,552$ [3]). Testing the adjacency-free predicate on every element returns a single positive: the state above, whose shortest word in $\{U^2, \ldots, R^2\}$ has length 6. Because the search is complete over a finite group, this is a proof, not a sample. Each of the six 180° turns swaps a face's four edge facelets with those of the opposite face, so each face ends up with its center color on one checkerboard class and the opposite color on the other; the counts are 9 per color. □



Pons Asinorum via $U^2D^2F^2B^2L^2R^2$ — the exact simulator output. Each face pairs a color with its opposite only; the color co-occurrence graph is the perfect matching $3K_2$.

Remark 1 (This is Construction A, made legal)**.** The "paired-faces" construction A of the companion note *is* realizable, precisely because it pairs each face with its *opposite* color, which is what fixed centers allow. Theorem 1 says it is the only checkerboard the mechanism permits.

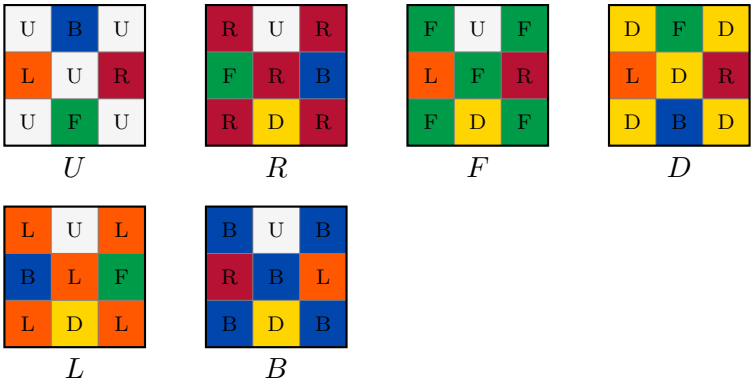
4 The full group: a second, different pattern exists

Theorem 2 (Non-uniqueness in the full group, verified)**.** *In the full Rubik group the adjacency-free property is* not *unique. The* ***superflip***

$$\boxed{U\,R^2\,F\,B\,R\,B^2\,R\,U^2\,L\,B^2\,R\,U'\,D'\,R^2\,F\,R'\,L\,B^2\,U^2\,F^2}$$

(all 12 *edges flipped in place;* 20 *moves, a known distance-*20 *state [4]) is adjacency-free and is* not *in the same symmetry class as Pons Asinorum. Hence there are at least two essentially different realizable adjacency-free patterns.*

Proof. The simulator applies the word above to the solved state and confirms the adjacency-free predicate and the per-color count 9. The two states are distinguished by the color co-occurrence graph (an isomorphism invariant under cube isometries and color relabelings, as shown in the companion note): for Pons it is $3K_2$ (each face shows only two colors, its own and its opposite), whereas for the superflip every face shows *five* colors — its own on the four corners and the center, and the four non-opposite neighbors on the edges — so the co-occurrence graph is the complete graph K_6. Since $3K_2 \ncong K_6$, the two patterns lie in different symmetry classes. □



Superflip — exact simulator output. Every face keeps its color on the four corners and center (an independent set in the grid) and carries its four edge-neighbors' colors on the edge facelets; no two orthogonal neighbors match. Co-occurrence graph K_6: distinct from Pons.

Proposition 1 (Many more exist)**.** *A* 4×10^6*-step random walk over the full group encountered* 28 distinct *adjacency-free states (all in the K_6 co-occurrence class, with* 4 *to* 6 *colors visible per face). Thus the realizable adjacency-free states number at least in the dozens — and surely far more — though their exact total is a separate hard enumeration problem (community estimates for related "no like-color adjacency" conditions run into the trillions [5]).*

Evidence. Direct output of the simulator's random walk; each state was reached by an explicit move sequence and verified adjacency-free. (A random walk under-counts and never reaches the measure-tiny half-turn subgroup, which is why Pons does not appear here; it is proved separately in Theorem 1.) □

5 Erratum to the companion counting note

The companion note ("On the Non-Uniqueness of Adjacency-Free Colorings") proved, for the *free-stickering* model, a lower bound $N(n) \geq \left((3n^2)!/(n^2!)^3\right)^2$, giving $\approx 1.2\times10^9$ at $n=2$ (the ordinary

cube). That theorem is correct *as a statement about free stickerings*, but two of its colorings do *not* survive the physical reachability constraint, and the headline number should not be read as a count of makeable patterns:

1. **The cyclic "Construction B" (C_6) is unrealizable.** It asks each face to show its own color paired with an *adjacent* (non-opposite) color. On a real cube the centers are fixed, so a face can only be checkerboarded against its *opposite* color (that is where the other stickers of that color live). Theorem 1 confirms Pons ($3K_2$) is the *only* two-color-per-face adjacency-free reachable state; no C_6 checkerboard exists on the physical cube.
2. **The disjoint-palette billion are mostly unreachable.** The 1.2×10^9 count permits any independent repainting of facelets within the parity classes. Real moves cannot repaint individual stickers — they move cubies carrying fixed color pairs/triples — so only a measure-zero fraction of those stickerings are reachable states.

Corrected takeaways.

- *Free stickering* (combinatorics): adjacency-free colorings are wildly non-unique — the companion bound stands.
- *Physical cube*, half-turn regime: adjacency-free is **unique** (Pons Asinorum), proven by exhaustive enumeration (Theorem 1).
- *Physical cube*, full group: adjacency-free is **not** unique (Pons and superflip are two different realizable patterns; dozens more exist), but the honest count is "at least two, empirically many," *not* a billion (Theorem 2, Prop. 1).

6 Verdict on "only one way"

Under the physical model the answer depends on the move set, and both halves are now proved rather than asserted:

$$\text{"only one way"} = \begin{cases} \textbf{TRUE} & \text{within the half-turn subgroup } \langle X^2 \rangle : \text{ Pons Asinorum, unique;} \\ \textbf{FALSE} & \text{in the full Rubik group: Pons, superflip, and more.} \end{cases}$$

The user's instinct — that there should be essentially one clean way to make such a pattern — is exactly right for the checkerboard family a cuber usually has in mind, and we can now say so with a complete finite proof. It is only when edge-flips (superflip and relatives) are admitted that a second family appears.

References

[1] *Rubik's Cube group*, Wikipedia. Order $|G| = 43{,}252{,}003{,}274{,}489{,}856{,}000 = 2^{27}3^{14}5^3 7^2 11$.

[2] D. Joyner, *Adventures in Group Theory: Rubik's Cube, Merlin's Machine, and Other Mathematical Toys*, 2nd ed., Johns Hopkins Univ. Press, 2008.

[3] The half-turn subgroup $\langle U^2, D^2, F^2, B^2, L^2, R^2 \rangle$ has order 663,552 (a Dihedral/elementary-abelian-by-permutation group); see the cube-theory literature and the squares-group analysis, e.g. [2].

[4] T. Rokicki, H. Kociemba, M. Davidson, J. Dethridge, *The Diameter of the Rubik's Cube Group Is Twenty*, SIAM J. Discrete Math. 27(2):1082–1105, 2013. (The superflip is a distance-20 position.)

[5] Speedsolving.com forum, *Number of cube states with no same-colored stickers adjacent* and *Superflip and checkerboards* (community discussions and estimates for "no like-color adjacency" conditions). Accessed 2026.